

```

import pandas as pd
import numpy as np
import nltk
import re

from nltk.corpus import stopwords, wordnet
from nltk.tokenize import word_tokenize

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

```

```

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
True

```

```

df = pd.read_csv("/content/job_descriptions.csv", engine='python', o

print(df.head())
print(df.columns)

```

```

/tmp/ipykernel_5950/3822726055.py:1: ParserWarning: Skipping line 43

```

```

df = pd.read_csv("/content/job_descriptions.csv", engine='python',

```

	Job Id	Experience	Qualifications	Salary Range	Location
0	1089843540111562	5 to 15 Years	M.Tech	\$59K-\$99K	Dubai
1	398454096642776	2 to 12 Years	BCA	\$56K-\$116K	As
2	481640072963533	0 to 12 Years	PhD	\$61K-\$104K	
3	688192671473044	4 to 11 Years	PhD	\$65K-\$91K	Port
4	117057806156508	1 to 12 Years	MBA	\$64K-\$87K	Sa

	Country	latitude	longitude	Work Type	Company Size
0	Isle of Man	54.2361	-4.5481	Intern	26801
1	Turkmenistan	38.9697	59.5563	Intern	100340
2	Macao SAR, China	22.1987	113.5439	Temporary	84525
3	Benin	9.3077	2.3158	Full-Time	129896
4	Chile	-35.6751	-71.5429	Intern	53944

	Contact	Job Title
0	001-381-930-7517x737	Digital Marketing Specialist
1	461-509-4216	Web Developer
2	9687619505	Operations Manager
3	+1-820-643-5431x47576	Network Engineer
4	343.975.4702x9340	Event Manager

	Role	Job Portal
0	Social Media Manager	Snagajob
1	Frontend Web Developer	Idealist
2	Quality Control Manager	Jobs2Careers

```

3 Wireless Network Engineer      FlexJobs
4       Conference Manager Jobs2Careers

```

```

                                Job Description \
0 Social Media Managers oversee an organizations...
1 Frontend Web Developers design and implement u...
2 Quality Control Managers establish and enforce...
3 Wireless Network Engineers design, implement, ...
4 A Conference Manager coordinates and manages c...

```

```

                                Benefits \
0 {'Flexible Spending Accounts (FSAs), Relocatio...
1 {'Health Insurance, Retirement Plans, Paid Tim...
2 {'Legal Assistance, Bonuses and Incentive Prog...
3 {'Transportation Benefits, Professional Develo...
4 {'Flexible Spending Accounts (FSAs), Relocatio...

```

```

                                skills \
0 Social media platforms (e.g., Facebook, Twitte...
1 HTML, CSS, JavaScript Frontend frameworks (e.g...
2 Quality control processes and methodologies St...
3 Wireless network design and architecture Wi-Fi...
4 Event planning Conference logistics Budget man...

```

```

                                Responsibilities \
0 Manage and grow social media accounts, create ...
1 Design and code user interfaces for websites, ...
2 Establish and enforce quality control standard...
3 Design, configure, and optimize wireless netwo...
4 Specialize in conference and convention planni...

```

```

df['content'] = (
    df['Job Title'].fillna('') + " " +
    df['Role'].fillna('') + " " +
    df['Job Description'].fillna('') + " " +
    df['skills'].fillna('') + " " +
    df['Responsibilities'].fillna('')
)
print(df[['Job Title', 'content']].head())

```

```

                                Job Title \
0 Digital Marketing Specialist
1       Web Developer
2       Operations Manager
3       Network Engineer
4       Event Manager

                                content
0 Digital Marketing Specialist Social Media Mana...
1 Web Developer Frontend Web Developer Frontend ...
2 Operations Manager Quality Control Manager Qua...
3 Network Engineer Wireless Network Engineer Wir...
4 Event Manager Conference Manager A Conference ...

```

```
stop_words = set(stopwords.words('english'))
```

```
def preprocess(text):
    text = text.lower()

    # remove special characters
    text = re.sub(r'[^a-zA-Z\s]', '', text)

    # tokenize
    tokens = word_tokenize(text)

    # remove stopwords
    tokens = [word for word in tokens if word not in stop_words]

    return " ".join(tokens)

df['cleaned_content'] = df['content'].apply(preprocess)

print(df[['content', 'cleaned_content']].head())
```

```

                                content \
0  Digital Marketing Specialist Social Media Mana...
1  Web Developer Frontend Web Developer Frontend ...
2  Operations Manager Quality Control Manager Qua...
3  Network Engineer Wireless Network Engineer Wir...
4  Event Manager Conference Manager A Conference ...

                                cleaned_content
0  digital marketing specialist social media mana...
1  web developer frontend web developer frontend ...
2  operations manager quality control manager qua...
3  network engineer wireless network engineer wir...
4  event manager conference manager conference ma...
```

#Task 2: TF-IDF

```
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df['cleaned_content'])
```

```
query = input("Enter job search query: ")
```

```
Enter job search query: Ai job
```

```
query_clean = preprocess(query)
```

```
query_vector = vectorizer.transform([query_clean])
```

```
similarity_scores = cosine_similarity(query_vector, tfidf_matrix)
```

```
scores = similarity_scores.flatten()
```

```
top_indices = scores.argsort()[-5:][::-1]
```

```
print("Top 5 Matching Jobs:\n")
```

```

for i in top_indices:
    print("Job Title:", df.iloc[i]['Job Title'])
    print("Company Name:", df.iloc[i]['Company'])
    print("Score:", scores[i])
    print("-"*50)

```

Top 5 Matching Jobs:

```

Job Title: HR Generalist
Company Name: Meta Platforms
Score: 0.121805145190331
-----
Job Title: HR Generalist
Company Name: Fidelity National Information Services
Score: 0.121805145190331
-----
Job Title: HR Generalist
Company Name: Sprint Corporation
Score: 0.121805145190331
-----
Job Title: HR Generalist
Company Name: Wesfarmers Limited
Score: 0.121805145190331
-----
Job Title: HR Generalist
Company Name: Petronet LNG
Score: 0.121805145190331
-----

```

#Task 3: Build Basic Search System

```

def basic_search(query):
    query_clean = preprocess(query)

    query_vector = vectorizer.transform([query_clean])

    similarity = cosine_similarity(query_vector, tfidf_matrix)

    scores = similarity.flatten()

    top_indices = scores.argsort()[-5:][::-1]

    print("\nTop 5 Search Results:\n")

    for i in top_indices:
        print("Job Title:", df.iloc[i]['Job Title'])
        print("Company Name:", df.iloc[i]['Company'])
        print("Location:", df.iloc[i]['location'])
        print("Score:", round(scores[i],4))
        print("-"*50)

```

#Task 4: Query Expansion using WordNet

```

def expand_query(query):
    expanded_terms = []

    for word in query.split():
        expanded_terms.append(word)

        synsets = wordnet.synsets(word)

        for syn in synsets[:2]:
            for lemma in syn.lemmas()[:2]:
                synonym = lemma.name().replace("_", " ")
                expanded_terms.append(synonym)

    expanded_query = " ".join(set(expanded_terms))

    return expanded_query
def search_with_expansion(query):
    expanded_query = expand_query(query)

    print("Original Query:", query)
    print("Expanded Query:", expanded_query)

    query_clean = preprocess(expanded_query)

    query_vector = vectorizer.transform([query_clean])

    similarity = cosine_similarity(query_vector, tfidf_matrix)

    scores = similarity.flatten()

    top_indices = scores.argsort()[-5:][::-1]

    print("\nTop 5 Results After Query Expansion:\n")

    for i in top_indices:
        print("Job Title:", df.iloc[i]['Job Title'])
        print("Company Name:", df.iloc[i]['Company'])
        print("Location:", df.iloc[i]['location'])
        print("Score:", round(scores[i],4))
        print("-"*50)

```

#Task 5 Result

```

print("WITHOUT QUERY EXPANSION")
basic_search("AI job")

print("\nWITH QUERY EXPANSION")
search_with_expansion("AI job")

```

WITHOUT QUERY EXPANSION

Top 5 Search Results:

Job Title: HR Generalist
Company Name: Meta Platforms
Location: Caracas
Score: 0.1218

Job Title: HR Generalist
Company Name: Fidelity National Information Services
Location: Ljubljana
Score: 0.1218

Job Title: HR Generalist
Company Name: Sprint Corporation
Location: Dili
Score: 0.1218

Job Title: HR Generalist
Company Name: Wesfarmers Limited
Location: Mbabane
Score: 0.1218

Job Title: HR Generalist
Company Name: Petronet LNG
Location: Gaza
Score: 0.1218

WITH QUERY EXPANSION

Original Query: AI job

Expanded Query: occupation business artificial intelligence Army Int

Top 5 Results After Query Expansion:

Job Title: Data Analyst
Company Name: Comcast
Location: Dushanbe
Score: 0.357

Job Title: Data Analyst
Company Name: Massachusetts Mutual Life Insurance
Location: Brasilia
Score: 0.357

Job Title: Data Analyst
Company Name: Aurobindo Pharma
Location: New Delhi
Score: 0.357

Job Title: Data Analyst
Company Name: ServiceNow
Location: Guatemala City
Score: 0.357

Job Title: Data Analyst
Company Name: Freeport-McMoRan